

Spatial figures. Polyhedra

The vocabulary of solids — face, edge, vertex — and the one formula that ties all three together.

LESSON 8–9

2 × 40 MIN

GEOMETRY 10, §5

IGCSE E4.8

LESSON CLOCK

10'

18'

22'

18'

20'

12'

The three words	10 min
Prisms and pyramids	18 min
Euler's formula	22 min
The five regular solids	18 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- Name the faces, edges and vertices of a polyhedron and count them.
- Verify Euler's formula for a range of solids.
- Distinguish convex from non-convex polyhedra.
- Name the five regular polyhedra and their face shapes.

TEXTBOOK

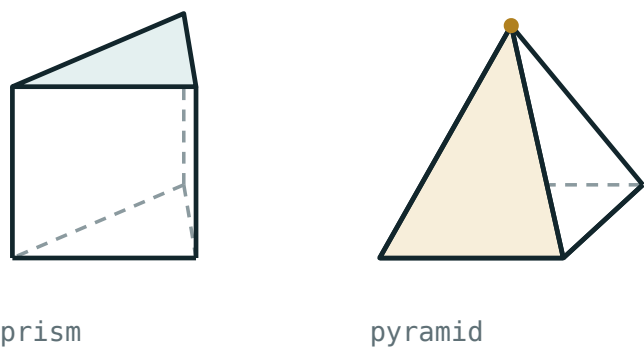
National	pp. 41–52
Cambridge	Core & Extended, pp. 282–288
Workbook	Exercise 5.1

01

Explanation

Face, edge, vertex

A **polyhedron** is a solid bounded by flat polygons. Those polygons are its **faces**; where two faces meet is an **edge**; where edges meet is a **vertex**.



A prism and a pyramid, with one face, one edge and one vertex marked on each.

SOLID	F	V	E
cube	6	8	12
triangular prism	5	6	9
square pyramid	5	5	8
tetrahedron	4	4	6
hexagonal prism	8	12	18
octahedron	8	6	12

Prisms and pyramids

TWO FAMILIES

A **prism** has two identical parallel bases joined by parallelograms. A **pyramid** has one base and all lateral edges meeting at a single apex.

For an n -gonal solid the counts follow from the base:

SOLID	F	V	E
n -gonal prism	$n + 2$	$2n$	$3n$
n -gonal pyramid	$n + 1$	$n + 1$	$2n$

A prism is **right** when the lateral edges are perpendicular to the bases, otherwise **oblique**. A pyramid is **regular** when its base is a regular polygon and the apex sits above the base centre.

Euler’s formula

FOR EVERY CONVEX POLYHEDRON

$F + V - E = 2$

Faces plus vertices, minus edges, is always two — whatever the solid, however many faces.

Check the table above: $6 + 8 - 12 = 2$, $5 + 6 - 9 = 2$, $4 + 4 - 6 = 2$. It never fails for a convex solid.

The formula is a **tool**, not a curiosity: it finds a missing count. A solid with 12 pentagonal faces has $E = \frac{12 \times 5}{2} = 30$ edges, so $V = 2 - F + E = 2 - 12 + 30 = 20$.

CONVEX ONLY

A polyhedron with a hole through it — a picture frame, say — gives $F + V - E = 0$. Euler’s formula assumes the surface can be deformed onto a sphere.

The five regular polyhedra

A polyhedron is **regular** when all faces are congruent regular polygons and the same number of them meet at every vertex. There are exactly five — a fact known to the Greeks:

NAME	FACES	<i>F</i>	<i>V</i>	<i>E</i>
tetrahedron	4 triangles	4	4	6
cube (hexahedron)	6 squares	6	8	12
octahedron	8 triangles	8	6	12
dodecahedron	12 pentagons	12	20	30
icosahedron	20 triangles	20	12	30

Why only five: at a vertex at least three faces must meet and their angles must total less than 360° . Triangles (60°) allow 3, 4 or 5; squares (90°) allow only 3; pentagons (108°) only 3; hexagons (120°) give exactly 360° and lie flat. Three plus one plus one — five solids, and no more are possible.

02 Worked examples**EXAMPLE 1** A polyhedron has 9 faces and 14 vertices. How many edges?

① $F + V - E = 2$

② $9 + 14 - E = 2$

③ $E = 21$

ANSWER 21 **EXAMPLE 2** A pentagonal prism: find F , V , E and check Euler.

① $n = 5: F = 7, V = 10, E = 15.$

② $7 + 10 - 15 = 2 \checkmark$

ANSWER $F = 7, V = 10, E = 15$

EXAMPLE 3 A solid has only triangular faces and 12 vertices, with 5 faces at each vertex. Find F and E .

- ① Each vertex has 5 edges; each edge has 2 ends.

$$E = \frac{12 \times 5}{2} = 30$$

- ② $F = 2 - V + E = 2 - 12 + 30 = 20$

- ③ This is the icosahedron.

ANSWER

$$F = 20, E = 30$$

03 Interactive model

Count F , V and E on a real box, then on a pyramid net, and check Euler each time.

Counting a polyhedron

A cube has how many edges?

A hexagonal pyramid has how many faces?

$F = 8$, $E = 12$ gives V equal to:

How many regular polyhedra are there?

3

4

5

infinitely many

A dodecahedron has faces that are:

triangles

squares

pentagons

hexagons

Why is there no regular solid with hexagonal faces?

angles sum to 360°

hexagons are not regular

too many edges

there is one

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Polyhedron	Ko'pyoq	Многогранник
Face	Yoq	Грань
Edge	Qirra	Ребро
Vertex	Uchi	Вершина
Convex polyhedron	Qavariq ko'pyoq	Выпуклый многогранник
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Regular polyhedron	Muntazam ko'pyoq	Правильный многогранник
Euler's formula	Eyler formulasi	Формула Эйлера
Lateral face	Yon yoq	Боковая грань
Base	Asos	Основание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Euler's formula is:

$$F + V + E = 2$$

$$F + V - E = 2$$

$$F - V + E = 2$$

$$F \cdot V = E$$

A triangular prism has:

$$F=5, V=6, E=9$$

$$F=6, V=5, E=9$$

$$F=5, V=9, E=6$$

$$F=4, V=4, E=6$$

A pyramid on an n -gon has how many edges?

$$n$$

$$n + 1$$

$$2n$$

$$3n$$

Where two faces meet is:

a vertex

an edge

a face

a base

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. F, V, E of a cube

ANSWER 6, 8, 12

2. F, V, E of a tetrahedron

ANSWER 4, 4, 6

3. F, V, E of a square pyramid

ANSWER 5, 5, 8

4. Check Euler for a cube

ANSWER $6 + 8 - 12 = 2$

5. Faces of an octahedron

ANSWER 8 triangles

6. Faces of a hexagonal prism

ANSWER 8

7. How many regular polyhedra?

ANSWER 5

Medium · the standard question

7 PROBLEMS

1. $F = 10, V = 16$; find E

ANSWER 24

2. $V = 12, E = 18$; find F

ANSWER 8

3. F, V, E of a heptagonal prism

ANSWER 9, 14, 21

4. F, V, E of a decagonal pyramid

ANSWER 11, 11, 20

5. A solid has 20 triangular faces. How many edges?

ANSWER 30

6. And how many vertices?

ANSWER 12

7. Which regular solid is that?

ANSWER icosahedron

Hard · stretch and reason

7 PROBLEMS

1. A solid has 12 pentagonal and 20 hexagonal faces, 3 at each vertex. Find V and E

ANSWER $V = 60, E = 90$

2. Show a polyhedron cannot have exactly 7 edges

ANSWER $F + V = 9$ forces an impossible face count

3. Prove $2E \geq 3F$ for any polyhedron

ANSWER Every face has at least 3 edges, each shared by 2 faces

4. Prove $2E \geq 3V$

ANSWER Every vertex meets at least 3 edges

5. Use both to show $F \leq 2V - 4$

ANSWER Substitute into Euler

6. A prism and a pyramid have the same number of edges. Relate their bases

ANSWER $3n = 2k$ — e.g. a hexagonal pyramid and a quadrilateral prism

7. Explain why a picture-frame solid fails Euler

ANSWER Its surface is a torus, not a sphere

07 Homework

Homework — 5 tasks

Task 5 needs the angle argument, not just the number five.

- Complete a table of F, V, E for a cube, a triangular prism, a pentagonal pyramid and an octahedron, and check Euler on each row.
- A polyhedron has 15 faces and 25 edges. Find its number of vertices.
- Write the formulas for F, V, E of an n -gonal prism and pyramid.
- Name the five regular polyhedra with their face shapes and counts.

5. Explain why no regular polyhedron can have hexagonal faces.